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### Magnetic disk apparatus and control method

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#### Abstract

In a magnetic disk apparatus according to an embodiment, a power supply circuit generates second power from first power supplied by an external power supply, and generates third power when the first power is cut off. The third power is generated based on regenerative energy generated by stoppage of a motor. A controller writes data received from a host device to a magnetic disk via a cache area by the second power while the first power is supplied, and executes a backup process when the first power is cut off. The backup process is executed by disabling communication with the host device and saving content of the cache area to the first memory by the third power. A monitoring circuit keeps connection with the external power supply even when the first power is cut off. The controller enables communication with the host device when the first power is restored.

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References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name   | U.S. Cl. | CPC          |
|--------------|-------------|-----------------|----------|--------------|
| 6295577      | 12/2000     | Anderson        | 714/24   | G06F 11/1441 |
| 7061707      | 12/2005     | Fong            | N/A      | G11B 21/12   |
| 8924641      | 12/2013     | Trantham et al. | N/A      | N/A          |
| 8947813      | 12/2014     | Chatradhi       | 714/24   | G11B 19/2081 |
| 9230593      | 12/2015     | Nicholls        | N/A      | G11B 19/20   |
| 9780700      | 12/2016     | Lieu            | N/A      | H02J 9/06    |
| 10635158     | 12/2019     | Watanabe        | N/A      | G06F 3/065   |
| 10656692     | 12/2019     | Kimura          | N/A      | G06F 3/0625  |
| 10996740     | 12/2020     | Watanbe et al.  | N/A      | N/A          |
| 11842047     | 12/2022     | Koyama          | N/A      | G06F 3/0619  |
| 2003/0028733 | 12/2002     | Tsunoda et al.  | N/A      | N/A          |
| 2015/0244207 | 12/2014     | Narita          | 307/64   | H02J 9/061   |
| 2019/0286208 | 12/2018     | Sogabe          | N/A      | G06F 11/3058 |
| 2020/0225733 | 12/2019     | Watanabe        | N/A      | G11B 19/2081 |
| 2023/0092423 | 12/2022     | Koyama          | N/A      | N/A          |

FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| H09-016338  | 12/1996          | JP      | N/A |
| H0916338    | 12/1996          | JP      | N/A |
| 2008-282137 | 12/2007          | JP      | N/A |
| 2008-282137 | 12/2007          | JP      | N/A |
| 5622289     | 12/2013          | JP      | N/A |
| 5622289     | 12/2013          | JP      | N/A |
| 2023-044277 | 12/2022          | JP      | N/A |
| 2023-044277 | 12/2022          | JP      | N/A |

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-035580, filed on Mar. 8, 2024; the entire contents of which are incorporated

herein by reference.

## FIELD

(2) Embodiments described herein relate generally to a magnetic disk apparatus and a control method.

## BACKGROUND

(3) A magnetic disk apparatus has a power loss protection (PLP) function. The PLP function is a function to, when supply of power from an external power supply is cut off, disable communication with a host device and save data, which is stored in a cache memory and is being written to a disk, to a nonvolatile storage area.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating an example of a configuration of a magnetic disk apparatus according to a first embodiment;
- (2) FIG. 2 is a diagram illustrating an example of a configuration of a magnetic disk apparatus illustrating a configuration of an SVC according to the first embodiment;
- (3) FIG. 3 is a diagram illustrating an example of a case where only a 5V power supply is restored in the configuration of the magnetic disk apparatus according to the first embodiment;
- (4) FIG. 4 is a flowchart illustrating an example of a procedure of a control process according to the first embodiment;
- (5) FIG. 5 is a flowchart illustrating an example of a procedure of a control process according to a second embodiment; and
- (6) FIG. 6 is a flowchart illustrating an example of a procedure (continuation) of the control process according to the second embodiment.

### DETAILED DESCRIPTION

(7) A magnetic disk apparatus according to an embodiment includes one or more magnetic disks, a motor, a power supply circuit, a first memory being non-volatile, a second memory being volatile and including a cache area, a controller, and a power supply monitoring circuit. The motor is configured to rotate the one or more magnetic disks. The power supply circuit is configured to generate second power from first power supplied by an external power supply, and generate third power when supply of the first power is cut off. The third power is generated based on regenerative energy generated by stoppage of the motor. The controller is configured to write data received from a host device to the magnetic disk via the cache area by using the second power generated by the power supply circuit while the first power is supplied. The controller is configured to execute a backup process when supply of the first power is cut off. The backup process is executed by disabling communication with the host device and saving content of the cache area to the first memory by using the third power generated by the power supply circuit. The power supply monitoring circuit is configured to monitor supply of the first power, and keep connection with the external power supply in a state where the supply of the first power is cut off. The controller is configured to enable communication with the host device in response to determining, by the power supply monitoring circuit, that the supply of the first power is restored.

(8) Hereinafter, a disc device and a control method according to embodiments will be described in detail with reference to the accompanying drawings. The present invention is not limited by the following embodiment.

#### First Embodiment

(9) FIG. 1 is a schematic diagram illustrating an example of a configuration of a magnetic disk apparatus 1 according to the embodiment. The magnetic disk apparatus 1 is connected to a host device 2. The magnetic disk apparatus 1 can receive an access command such as a write command

or a read command from the host device **2**.

(10) The magnetic disk apparatus **1** includes a magnetic disk **11** having a magnetic layer formed on a face thereof. The magnetic disk apparatus **1** accesses the magnetic disk **11** in response to the access command. The access includes writing of data and reading of data.

(11) Data is written and read via a magnetic head **22**. The magnetic disk apparatus **1** includes, in addition to the magnetic disk **11**, a spindle motor (SPM) **12**, a ramp **13**, an actuator arm **15**, a voice coil motor (VCM) **16**, a servo controller (SVC) **21**, a magnetic head **22**, a hard disk controller (HDC) **23**, a host interface (host I/F) **31**, a preamplifier **24**, a read/write channel (RWC) **25**, a processor **26**, sensors **27**, a flash read only memory (FROM) **28**, a dynamic random access memory (DRAM) **29**, a 12V eFuse **41**, a 5V eFuse **43**, and a PLP regulator **42**.

(12) The magnetic disk **11** is rotated at a predetermined rotation speed by an SPM **12** attached coaxially.

(13) The SVC **21** is an integrated circuit having a function as a driver that drives the SPM **12** and the VCM **16**. Through the SVC **21**, the processor **26** controls the rotation of the SPM **12** and the rotation of the VCM **16**. Details of the SVC **21** will be described later.

(14) The magnetic head **22** performs information writing and reading on the magnetic disk **11** by a write head **22w** and a read head **22r** provided in the magnetic head **22**. The magnetic head **22** is fixed to a distal end of the actuator arm **15**. The magnetic head **22** is moved in the radial direction of the magnetic disk **11** by the VCM **16** driven by the SVC **21**. Note that one or both of the write head **22w** and the read head **22r** included in the magnetic head **22** may be provided in multiple numbers for a single magnetic head **22**.

(15) When, for example, the rotation of the magnetic disk **11** is stopped, the magnetic head **22** is moved onto the ramp **13**. The ramp **13** is configured to hold the magnetic head **22** at a position spaced apart from the magnetic disk **11**.

(16) The preamplifier **24** is an integrated circuit that writes and reads data via the magnetic head **22**. The preamplifier **24** amplifies and outputs a signal read from the magnetic disk **11** by the magnetic head **22** during the read operation, and supplies the signal to the RWC **25**. In addition, the preamplifier **24** amplifies a signal corresponding to the data to be written supplied by the RWC **25** and supplies the amplified signal to the magnetic head **22** during the write operation.

(17) The host I/F **31** is a communication interface with the host device **2**. The HDC **23** controls transmission and reception of data with the host device **2** via the host I/F **31**, and controls the DRAM **29**.

(18) The DRAM **29** is used as a buffer for data to be transmitted to and received from the host device **2**. Thus, the cache area **291** is allocated to the DRAM **29**, and the DRAM **29** functions as a cache memory. The cache area **291** is used for temporarily storing data to be written that is not yet written to the magnetic disk **11** out of data received from the host device **2**. The DRAM **29** is also used for temporarily storing data read from the magnetic disk **11**.

(19) The DRAM **29** is used as an operation memory by the processor **26**. The DRAM **29** is used as an area where firmware is loaded and an area where various types of management data are temporarily stored. The DRAM **29** is an example of the second memory being a volatile memory.

(20) The RWC **25** performs code modulation on data to be written, which is supplied by the HDC **23** and stored in the cache area **291**, and supplies the code-modulated data to the preamplifier **24**. The RWC **25** also performs code demodulation including error correction on a signal that is read from the magnetic disk **11** and supplied by the preamplifier **24**, and outputs the code-demodulated signal to the HDC **23** as digital data.

(21) In one example, the processor **26** is a central processing unit (CPU). The sensors **27**, the flash read only memory (FROM) **28**, and the DRAM **29** are connected to the processor **26**.

(22) The FROM **28** stores firmware (program data), various operation parameters, and the like. The firmware may be stored in the magnetic disk **11**. The FROM **28** is an example of the first memory being a non-volatile memory.

(23) The sensors **27** are sensor modules that detect environment information. The sensors **27** include, for example, an acceleration sensor module for detecting vibration or impact applied to the magnetic disk apparatus **1**, a temperature sensor module for detecting the temperature of the magnetic disk apparatus **1**, etc.

(24) The processor **26** performs overall control of the magnetic disk apparatus **1** according to a firmware stored in the FROM **28** or the magnetic disk **11**. For example, the processor **26** loads a firmware from the FROM **28** or the magnetic disk **11** into the DRAM **29**, and controls, according to the firmware loaded into the DRAM **29**, the SVC **21**, the preamplifier **24**, the RWC **25**, the HDC **23**, and the like.

(25) The HDC **23**, the host I/F **31**, the RWC **25**, and the processor **26** are configured as a system-on-a-chip (SoC) **30** that is one integrated circuit. In addition to these element, the SoC **30** may include other elements (for example, the FROM **28**, the DRAM **29**, or the like).

(26) Some of or all the functions of the processor **26** may be implemented by a hardware circuit such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC).

(27) FIG. **2** is a diagram illustrating an example of a configuration of the magnetic disk apparatus **1** illustrating a configuration of the SVC **21** according to the first embodiment.

(28) An I/F connector **45** is a general term for various terminals, and is provided in a housing of the magnetic disk apparatus **1**. The host device **2** is provided with a 12V power supply and a 5V power supply as external power supplies. The 12V power supply is an example of a first external power supply, and the 5V power supply is an example of a second external power supply.

(29) The 12V eFuse **41** (an example of a first electronic fuse) is connected to the 12V power supply via the I/F connector **45**. The I/F connector **45** to which the 12V eFuse **41** is connected is a terminal for 12V power supply. The 12V eFuse **41** is also connected to the SVC **21**. Power supplied by the 12V power supply is input to the 12V eFuse **41** via the I/F connector **45** (namely, the terminal for 12V power supply).

(30) The 12V eFuse **41** is an integrated circuit that include a MOSFET and has an overcurrent protection function to prevent an overcurrent from flowing through a power supply lane on which the 12V eFuse **41** is provided. The overcurrent protection function enables the 12V eFuse **41** to perform detection of overcurrent and perform cutoff of power when overcurrent is detected. When the 12V power supply is cut off, the 12V eFuse **41** is turned off. By being turned on, the 12V eFuse **41** supplies the SVC **21** with power that is supplied by the 12V power supply when the 12V power supply is restored.

(31) The 5V eFuse **43** (an example of a second electronic fuse) is connected to a 5V power supply via the I/F connector **45**. The I/F connector **45** to which the 5V eFuse **43** is connected is a terminal for 5V power supply. The 5V eFuse **43** is also connected to the SVC **21**. Power supplied by the 5V power supply is input to the 5V eFuse **43** via the I/F connector **45** (namely, the terminal for 5V power supply).

(32) The 5V eFuse **43** is an integrated circuit that includes a MOSFET and has an overcurrent protection function to prevent an overcurrent from flowing through a power supply lane on which the 5V eFuse **43** is provided. The overcurrent protection function enables the 5V eFuse **41** to perform detection of overcurrent and perform cutoff of power when overcurrent is detected. When the 5V power supply is cut off, the 5V eFuse **43** is turned off. By being turned on, the 5V eFuse **43** supplies the SVC **21** with power that is supplied by the 5V power supply when the 5V power supply is restored.

(33) The PLP regulator **42** supplies power to the SVC **21** for a PLP function to be described later. When the supply of power from the 12V power supply is cut off, the PLP regulator **42** supplies a back electromotive force of the SPM **12**, which is supplied by a spindle motor control circuit **212** to be described later. If the supply of power from the 12V power supply is not cut off, power (an example of the second power) is generated based on power supplied by the 12V power supply via

the FET **211**, and the generated power is supplied to the SVC **21**. Here, generation of power includes, for example, processing such as rectification, boosting voltage, or reducing voltage. The PLP regulator **42** may directly supply the supplied power to each unit.

(34) When the supply of power from the external power supply (12V power supply and/or 5V power supply) provided in the host device **2** is cut off, a power supply monitoring circuit **213** to be described later can detect that the supply of power from the host device **2** is cut off. In a case where the power supply monitoring circuit **213** detects that the supply of power from the host device **2** is cut off, the PLP regulator **42** receives regenerative energy generated by the SPM **12** via the spindle motor control circuit **212**. Then, the PLP regulator **42** generates power based on the regenerative energy from the SPM **12** and supplies the generated power to the SVC **21**. In other words, the PLP regulator **42** generates power based on the back electromotive force from the SPM **12** and supplies the generated power to the SVC **21**.

(35) The power supplied by the 12V power supply is an example of the first power and the fourth power. The power supplied by the 5V power supply is an example of the first power and the fifth power. The power generated based on the back electromotive force from the SPM **12** is an example of the third power.

(36) When the current supplied to the SPM **12** is stopped during the rotation of the magnetic disk **11**, the energy caused by the back electromotive force of the SPM **12** is recovered as regenerative energy by the spindle motor control circuit **212**. In addition, kinetic energy of the rotating magnetic disk is converted into electric energy by the SPM **12**. The electric energy is recovered as regenerative energy by the spindle motor control circuit **212**.

(37) As described above, the PLP regulator **42** generates power to be supplied to each unit, based on regenerative energy that is generated by stoppage of the SPM **12**. Therefore, the magnetic disk apparatus **1** can operate by using the regenerative energy for a while after the supply of power from the host device **2** is cut off. In this period of time, the PLP function processing is performed by the magnetic disk apparatus **1**.

(38) The PLP function is a function to save, to the nonvolatile memory, data stored in the cache memory of the DRAM **29** in order to prevent the data from being lost from the magnetic disk apparatus **1** when the cut-off of the supply of power from the host device **2** is detected while the data is written to the magnetic disk **11**. In the case of the present embodiment, data in the cache area **291** is saved to the FROM **28**, as a process of the PLP function.

(39) Here, the wording of “saving data” refers to processing of recording the data temporarily in another recording area. As long as the data is recorded temporarily in another recording area, the “saving data” can be interpreted as transferring the data, copying the data, or the like. The saving may include a process of processing data, such as compression, encoding, format conversion, and the like. Hereinafter, a process of saving data in the cache area **291** to the FROM **28** may be referred to as a backup process.

(40) As illustrated in FIG. **2**, the SVC **21** mainly includes an FET **211**, a spindle motor control circuit **212**, a power supply monitoring circuit **213**, a 1.8V regulator **214**, a 1.5V regulator **215**, and a 0.9V regulator **216**.

(41) The FET **211** is a field effect transistor and is connected between the spindle motor control circuit **212** and the 12V eFuse **41**. When turned on in accordance with an on/off control signal from the power supply monitoring circuit **213**, the FET **211** supplies the spindle motor control circuit **212** with power that is supplied by the 12V power supply. When turned off in accordance with an on/off control signal from the power supply monitoring circuit **213**, power supplied by the 12V power supply is cut off. The spindle motor control circuit **212** controls the rotation of the SPM **12**.

(42) The power supply monitoring circuit **213** monitors supply of power from the host device **2** by monitoring power supplied from the 12V power supply and power supplied from the 5V power supply. The power supply monitoring circuit **213** is connected to the 12V eFuse **41**. The power supply monitoring circuit **213** monitors the power supply voltage of the 12V power supply, and

monitors whether the 12V power supply is in the power interruption state or is restored from the power interruption state. The power supply monitoring circuit **213** is also connected to the 5V eFuse **43**. The power supply monitoring circuit **213** monitors a power supply voltage of the 5V power supply, and monitors whether the 5V power supply is in a power interruption state or is restored from the power interruption state.

(43) In response to detecting that the supply of power from the host device **2** is cut off, the power supply monitoring circuit **213** turns off the FET **211** by negating the on/off control signal output to the FET **211**. In addition, the power supply monitoring circuit **213** asserts an on/off control signal for the PLP regulator **42** in response to detecting that the supply of power from the host device **2** is cut off. As a result, the PLP function can be backed up. Hereinafter, cut-off of supply of power from the host device **2** may be referred to as power loss.

(44) The 1.8V regulator **214** generates power of 1.8 V based on the power from the 12V eFuse **41**, the 5V eFuse **43**, and the PLP regulator **42**, and supplies the generated power to the FROM **28**.

(45) The 1.5V regulator **215** generates power of 1.5 V based on the power from the 12V eFuse **41**, the 5V eFuse **43**, and the PLP regulator **42**, and supplies the generated power to the DRAM **29**.

(46) The 0.9V regulator **216** generates power of 0.9 V based on the power from the 12V eFuse **41**, the 5V eFuse **43**, and the PLP regulator **42**, and supplies the generated power to the SoC **30**.

(47) The PLP regulator **42**, the 1.8V regulator **214**, the 1.5V regulator **215**, and the 0.9V regulator **216** are each an example of a power supply circuit.

(48) The HDC **23** is an example of a controller. The HDC **23** may include a central processing unit (CPU), a logic circuit, or both. The HDC **23** controls the entire magnetic disk apparatus **1** according to a command received from the host device **2**.

(49) The HDC **23** executes a normal operation using the power supplied by the 0.9V regulator **216** while receiving supply of power from the host device **2**. The normal operation includes transmission and reception of commands and data via the host I/F **31** to and from the host device **2**, and access to (write to and read from) the magnetic disk **11**.

(50) In the normal operation, when data requested to be written by the write command is received from the host device **2**, the HDC **23** stores the received data in the cache area **291**. The HDC **23** causes the RWC **25** to write the data in the cache area **291** to the magnetic disk **11**. In other words, the HDC **23** writes the data received from the host device **2** to the magnetic disk **11** via the cache area **291**. The HDC **23** transmits data output from the RWC **25** to the host device **2**.

(51) When the external power supply is in a power interruption state and the power supply is cut off (namely, when power loss occurs), the HDC **23** terminates the normal operation and disables the communication with the host device **2** by negating the host I/F enable signal.

(52) The host I/F enable signal is a signal for instructing the host I/F **31** to enable/disable communication. By asserting the host I/F enable signal, the host I/F **31** is enabled, and communication with the host device **2** is enabled. On the other hand, by negating the host I/F enable signal, the host I/F **31** is invalid, and communication with the host device **2** is invalid. Here, the host I/F enable signal is a signal for controlling communication with the host device **2**. By assertion of the host I/F enable signal, communication with the host device **2** is enabled, and communication with the host device **2** is disabled by negation.

(53) Then, the HDC **23** executes a backup process related to the PLP function. The power supply monitoring circuit **213** makes a notification of the cut-off of the power supply (namely, the occurrence of the power loss) due to the cut-off of the external power supply. The HDC **23** can execute the backup process in a period during which the backup enable signal is asserted after receiving the notification of the cut-off of the power supply. The HDC **23** executes the backup process by using the power supplied by the PLP regulator **42**, namely, using the power generated from the regenerative energy that is generated by stoppage of the SPM **12**.

(54) In the present embodiment, the HDC **23** keeps connection between the external power supply and the magnetic disk apparatus **1** even when the power supply is cut off. Then, the HDC **23**

enables communication with the host device **2** when the power supply monitoring circuit **213** determines that the power supply is restored.

(55) More specifically, even when the power supply monitoring circuit **213** detects that either one or both of the supply of power from the 12V power supply and the supply of power from the 5V power supply are cut off, the HDC **23** keeps connection with the 12V power supply and connection with the 5V power supply, and starts the backup process. In a state where supply of power from the 12V power supply alone is cut off, or in a state where supply of power from the 5V power supply alone is cut off, or in a state where supply of power from both the 12V power supply and the 5V power supply is cut off, the HDC **23** turns on the 12V eFuse **41** and the 5V eFuse **43**, and starts the backup process. Therefore, the power supply monitoring circuit **213** can detect restoration of the 12V power supply and restoration of the 5V power supply.

(56) In the example of FIG. **2** depicting the 12V eFuse **41** being turned on, the 12V power supply is restored. Alternatively, the 5V power supply may be restored in some cases.

(57) FIG. **3** is a diagram illustrating an example of a case where only a 5V power supply is restored in the configuration of the magnetic disk apparatus **1** according to the first embodiment. In the example of FIG. **3** depicting the 5V eFuse **43** being turned on, the 5V power supply is restored. Alternatively, both the 12V power supply and the 5V power supply may be restored in some cases.

(58) As described above, in the present embodiment, the connection with the 12V power supply and the connection with the 5V power supply are kept by the turned-on 12V eFuse **41** and the turned-on 5V eFuse **43**, respectively, during the backup process. Therefore, when the supply of power from the 12V power supply or the supply of power from the 5V power supply is restored, the power supply monitoring circuit **213** can detect their restoration. When the power supply monitoring circuit **213** detects that the supply of power from the 12V power supply and/or the supply of power from the 5V power supply, which were/was cut off but are/is thereafter restored, the HDC **23** asserts the host I/F enable signal to enable the communication with the host device **2**.

(59) The communication with the host device **2** is enabled during the backup process, so that there is a case where write data is transmitted from the host device **2**. In such a case, the HDC **23** receives the write data transmitted from the host device **2** during the backup process, and stores the received write data in the FROM **28**.

(60) Next, the control process by the magnetic disk apparatus **1** according to the present embodiment configured as described above will be described. FIG. **4** is a flowchart illustrating an example of a procedure of a control process according to the first embodiment. It is assumed that the external power supply (namely, the 5V power supply or the 12V power supply) is in a power interruption state while the magnetic disk apparatus **1** is executing the normal process (S11). In this case, the power supply monitoring circuit **213** detects Fault and determines that the supply of power from the 5V power supply or the 12V power supply is cut off (S12). Then, the power supply monitoring circuit **213** turns on the 12V eFuse **41** and the 5V eFuse **43** (S13).

(61) Next, the power supply monitoring circuit **213** asserts an on/off control signal for the PLP regulator **42**, whereby the processing is switched to the execution of the PLP function (S14). Thereafter, the HDC **23** disables communication with the host device **2** (S15). Subsequently, the power supply monitoring circuit **213** determines whether the power supply is restored by restoration of the external power supply (12V power supply or 5V power supply) (S16). In response to determining that the external power supply is not restored (S16: No), the PLP regulator **42** generates power of 5 V from the back electromotive force of the SPM **12** and supplies the generated power of 5 V to the SVC **21** (S17). The HDC **23** executes the operation of the PLP function and executes the backup process (S18). Thereafter, the process proceeds to S25.

(62) On the other hand, in response to determining that the external power supply (12V power supply or 5V power supply) is restored and the supply of power from the external power supply is also restored (S16: Yes), the PLP regulator **42** generates power of 5 V from the power of the restored external power supply (12V power supply or 5V power supply) and supplies the generated



power of 5 V to the SVC **21** (S19).

(63) Next, the HDC **23** enables communication with the host device **2** (S20). The HDC **23** determines whether data backup is necessary (S21). Specifically, the HDC **23** determines whether the operation mode of the magnetic disk apparatus **1** is a standby state, an idle B state, or an idle C state.

(64) The idle B state refers to a state where the SPM **12** normally rotates (for example, 7200 rpm) while the magnetic head **22** is retracted. The idle C state is a state where the SPM **12** rotates at a low speed (for example, 3200 rpm) in a state where the magnetic head **22** is retracted.

(65) When the operation mode of the magnetic disk apparatus **1** is any of the standby state, the idle B state, and the idle C state, backup of data is not necessary (S21: No). In this case, the process proceeds to S27, the HDC **23** restarts the magnetic disk apparatus **1** (S27), and the process ends.

(66) If the operation mode of the magnetic disk apparatus **1** is not the standby state, the idle B state, or the idle C state, backup of data is necessary (S21: Yes). In this case, the HDC **23** executes the operation of the PLP function and executes the backup process (S22).

(67) Next, the HDC **23** determines whether a write command has been received from the host device **2** (S23). In response to determining that the write command has been received from the host device **2** (S22: Yes), the write data is stored in the FROM **28** (S24). In response to determining that the write command is not yet received from the host device **2** (S23: No), the processing of S24 is not performed.

(68) Next, the HDC **23** determines whether the PLP operation has timed out (S25). In response to determining that the PLP operation has not timed out (S25: No), the process returns to S22, and the process from S22 to S24 is repeatedly executed.

(69) On the other hand, in response to determining that the PLP operation has timed out (S25: Yes), the HDC **23** completes the PLP operation (S26). Then, the HDC **23** restarts the magnetic disk apparatus **1** (S27), and the process ends.

(70) As a side note, in a case of a magnetic disk apparatus of a comparative example, when the supply of power from an external power supply is interrupted during the operation, a backup process of writing data of DRAM to FROM is performed by using the back electromotive force of SPM by the PLP function. At this time, in order to maintain the back electromotive force, unnecessary circuits and unnecessary communication interfaces are disabled. However, in such a case, the magnetic disk apparatus of the comparative example cannot be restarted and communication with a host device cannot be performed until the PLP function times out. Therefore, if duration of a timer of the host device is short, the host device cannot recognize the magnetic disk apparatus due to timeout.

(71) Considering the above issue, in the magnetic disk apparatus **1** according to the present embodiment, the power supply monitoring circuit **213** keeps the connection with the external power supply (12V power supply and 5V power supply) even when the supply of power from the external power supply is cut off. Then, when the power supply monitoring circuit **213** determines that the supply of power from the external power supply is restored, the HDC **23** enables the communication with the host device **2**.

(72) Therefore, according to the present embodiment, even when the PLP function is executed due to cutoff of supply of power from the external power supply and the backup process is being executed, the restoration of the external power supply can be monitored, and the communication with the host device **2** can be enabled immediately when the restoration of the external power supply is detected. Therefore, according to the present embodiment, it is possible to avoid the occurrence of timeout by the host device **2** due to the disconnection between the host device **2** and the magnetic disk apparatus **1**, and possible to smoothly communicate with the host device **2** after restoration of the external power supply.

(73) In addition, in the magnetic disk apparatus **1** according to the present embodiment, the power supply monitoring circuit **213** monitors the supply of power from the 12V power supply and the

supply of power from the 5V power supply, and keeps connection between the 12V power supply and the magnetic disk apparatus **1** and also keeps connection between the 5V power supply and the magnetic disk apparatus **1** even when either one or both of the supply of power from the 12V power supply and the supply of power from the 5V power supply are cut off. Thereafter, the HDC **23** enables communication with the host device **2** when the power supply monitoring circuit **213** determines that either the supply of power from the 12V power supply or the supply of power from the 5V power supply is restored.

(74) Therefore, according to the present embodiment, even when the PLP function is executed due to cutoff of the supply of power from the 12V power supply or the supply of power from the 5V power supply and the backup process is being executed, the restoration of the supply of power from the 12V power supply and the supply of power from the 5V power supply can be monitored, and the communication with the host device **2** can be enabled immediately when the restoration of the external power supply is detected. Therefore, according to the present embodiment, it is possible to avoid the occurrence of timeout by the host device **2** due to the disconnection between the host device **2** and the magnetic disk apparatus **1**, and possible to smoothly communicate with the host device **2** after restoration of the external power supply.

(75) Moreover, in the magnetic disk apparatus **1** according to the present embodiment, the power supply monitoring circuit **213** turns on the 12V eFuse **41** connected between the 12V power supply and the power supply monitoring circuit **213** and the 5V eFuse **43** connected between the 5V power supply and the power supply monitoring circuit **213** even when either one or both of the supply of power from the 12V power supply and the supply of power from the 5V power supply are cut off.

(76) Therefore, according to the present embodiment, even when the PLP function is executed due to cutoff of the supply of power from the 12V power supply or the supply of power from the 5V power supply and the backup process is being executed, it is possible to reliably monitor the restoration of the supply of power from the 12V power supply and the supply of power from the 5V power supply. Therefore, according to the present embodiment, after the 12V power supply or the 5V power supply that was cut off is restored, it is possible to more smoothly communicate with the host device **2** by avoiding the occurrence of timeout by the host device **2** due to the disconnection between the host device **2** and the magnetic disk apparatus **1**.

(77) In the magnetic disk apparatus **1** according to the present embodiment, when the communication with the host device **2** is enabled during the backup process of the PLP function, the HDC **23** receives the write data transmitted from the host device **2** and stores the received write data in the FROM **28**.

(78) Therefore, according to the present embodiment, when the 12V power supply or the 5V power supply that was cut off is restored, a write command can be transmitted from the host device **2** to the magnetic disk apparatus **1** while avoiding the occurrence of timeout by the host device **2**. Thus, communication with the host device **2** can be more smoothly performed.

## Second Embodiment

(79) In the above-described first embodiment, even when the external power supply is interrupted, the magnetic disk apparatus **1** keeps the connection with the external power supply. Thereafter, when the external power supply is restored, the magnetic disk apparatus **1** enables the communication with the host device **2**, and writes data to the FROM **28** when the write command is received from the host device **2** during the PLP operation. In the second embodiment that will be described, information about the state of the magnetic disk apparatus **1**, such as the progress of the PLP operation, is transmitted to the host device **2**.

(80) The configuration of the magnetic disk apparatus **1** according to the present embodiment is similar to that of the first embodiment illustrated in FIGS. **1** to **3**.

(81) In addition to having a function similar to that of the first embodiment, the HDC **23** according to the present embodiment transmits information about the state of the magnetic disk apparatus **1** to the host device **2** when the communication with the host device **2** is enabled during execution of

the backup process by the PLP function.

(82) The information about the state of the magnetic disk apparatus **1** includes at least the progress status of the backup process by the PLP function and/or the operation mode of the magnetic disk apparatus **1**.

(83) Moreover, upon receiving an instruction to suspend the backup process from the host device **2** after transmitting the information about the state of the magnetic disk apparatus **1** to the host device **2**, the HDC **23** according to the present embodiment terminates the backup process of the PLP function before a timeout of the backup process of the PLP function.

(84) Next, the control process by the magnetic disk apparatus **1** according to the present embodiment configured as described above will be described. FIGS. **5** and **6** are flowcharts illustrating an example of a procedure of a control process according to the second embodiment.

(85) Processing (S**11** to S**24**) from the occurrence of the 5V power supply interruption or the 12V power supply interruption to the reception of the write command and the storage of the write data is executed as in the first embodiment.

(86) After S**24**, the HDC **23** transmits the progress status of the PLP operation to the host device **2** as information about the state of the magnetic disk apparatus **1** (S**31**). The HDC **23** may transmit the operation mode of the magnetic disk apparatus **1** to the host device **2**.

(87) Next, the HDC **23** determines whether an instruction to interrupt the PLP operation (namely, the backup process) has been received from the host device **2** (S**32**). In response to determining that the instruction to interrupt the PLP operation (backup process) is not yet received from the host device **2** (S**32**: No), the process proceeds to S**25**, and the process similar to that of the first embodiment is performed.

(88) On the other hand, in response to determining that the instruction to interrupt the PLP operation is received from the host device **2** (S**32**: Yes), the HDC **23** interrupts the PLP operation without waiting for a timeout of the PLP operation. Then, the HDC **23** restarts the magnetic disk apparatus **1** (S**27**), and the process ends.

(89) As described above, in the magnetic disk apparatus **1** according to the present embodiment, when the communication with the host device **2** is enabled during the backup process of the PLP function, the HDC **23** transmits the information about the state of the magnetic disk apparatus **1** to the host device **2**.

(90) Therefore, even when the PLP function is executed due to cutoff of the supply of power from the external power supply and the backup process is being executed, information about the state of the magnetic disk apparatus **1** is transmitted to the host device **2** when the cut off external power supply is restored. Thus, it is possible to avoid occurrence of timeout by the host device **2** due to disconnection between the host device **2** and the magnetic disk apparatus **1**, and possible to more smoothly communicate with the host device **2**.

(91) Moreover, in the magnetic disk apparatus **1** according to the present embodiment, the information about the state of the magnetic disk apparatus **1** includes at least the progress status of the backup process of the PLP function and/or the operation mode of the magnetic disk apparatus **1**.

(92) Therefore, according to the present embodiment, even when the PLP function is executed due to cutoff of the supply of power from the external power supply and the backup process is being executed, at least the progress status of the backup process of the PLP function and/or the operation mode of the magnetic disk apparatus **1** is transmitted to the host device **2** when the cut off external power supply is restored. Therefore, it is possible to avoid occurrence of timeout by the host device **2** due to disconnection between the host device **2** and the magnetic disk apparatus **1**, and possible to more smoothly communicate with the host device **2**.

(93) In addition, in the magnetic disk apparatus **1** according to the present embodiment, when an instruction to interrupt the backup process is received from the host device **2** after transmitting the information about the state of the magnetic disk apparatus **1** to the host device **2**, the HDC **23** terminates the backup process as the PLP operation without waiting for the timeout of the backup

process.

(94) Therefore, according to the present embodiment, in a case where the PLP function is executed due to cutoff of the supply of power from the external power supply and the cut off external power supply is thereafter restored during the backup process being executed, the backup process is terminated without waiting for a timeout of the backup process when an instruction to interrupt the backup process is received from the host device 2. Therefore, the communication with the host device 2 can be performed more smoothly.

(95) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; moreover, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

## Claims

1. A magnetic disk apparatus comprising: one or more magnetic disks; a motor configured to rotate the one or more magnetic disks; a power supply circuit configured to generate second power from first power supplied by an external power supply, and generate third power when supply of the first power is cut off, the third power being generated based on regenerative energy generated by stoppage of the motor; a first memory being non-volatile; a second memory being volatile and including a cache area; a controller configured to write data received from a host device to the one or more magnetic disks via the cache area by using the second power generated by the power supply circuit while the first power is supplied, and execute a backup process when supply of the first power is cut off, the backup process being executed by disabling communication with the host device and saving content of the cache area to the first memory by using the third power generated by the power supply circuit; and a power supply monitoring circuit configured to monitor supply of the first power, and keep connection with the external power supply in a state where the supply of the first power is cut off, wherein the controller is configured to enable communication with the host device in response to determining, by the power supply monitoring circuit, that the supply of the first power is restored.

2. The magnetic disk apparatus according to claim 1, wherein the external power supply includes a first external power supply configured to supply fourth power as the first power, and a second external power supply configured to supply fifth power as the first power, the power supply monitoring circuit is configured to monitor supply of the fourth power and supply of the fifth power, and keep connection with the first external power supply and connection with the second external power supply in a state where either one or both of the supply of the fourth power and the supply of the fifth power are cut off; and the controller is configured to enable communication with the host device in response to determining, by the power supply monitoring circuit, that either the cut off supply of the fourth power or the cut off supply of the fifth power is restored.

3. The magnetic disk apparatus according to claim 2, further comprising: a first electronic fuse connected between the first external power supply and the power supply circuit, and between the first external power supply and the power supply monitoring circuit, the first electronic fuse being configured to enable, by being turned on, supply of the fourth power from the first external power supply to the power supply circuit, and cut off the supply of the fourth power by being turned off; and a second electronic fuse connected between the second external power supply and the power supply circuit, and between the second external power supply and the power supply monitoring circuit, the second electronic fuse being configured to enable, by being turned on, supply of the fifth power from the second external power supply to the power supply circuit, and cut off the

- supply of the fifth power by being turned off, wherein the power supply monitoring circuit is configured to turn on the first electronic fuse and the second electronic fuse in a state where either one or both of the supply of the fourth power and the supply of the fifth power are cut off.
4. The magnetic disk apparatus according to claim 1, wherein the controller is configured to receive write data transmitted from the host device and store the received write data in a first memory when communication with the host device is enabled during execution of the backup process.
5. The magnetic disk apparatus according to claim 1, wherein the controller is configured to transmit information about a state of the magnetic disk apparatus to the host device when communication with the host device is enabled during execution of the backup process.
6. The magnetic disk apparatus according to claim 5, wherein the information about the state of the magnetic disk apparatus includes a progress status of the backup process and/or an operation mode of the magnetic disk apparatus.
7. The magnetic disk apparatus according to claim 5, wherein the controller is configured to terminate the backup process without waiting for a timeout of the backup process in response to receiving an instruction to interrupt the backup process from the host device after transmitting the information about the state of the magnetic disk apparatus to the host device.
8. A control method implemented by a magnetic disk apparatus, the magnetic disk apparatus including one or more magnetic disks, a motor serving to rotate the one or more magnetic disks, a power supply circuit serving to generate second power from first power supplied by an external power supply and generate third power when supply of the first power is cut off, the third power being generated based on regenerative energy generated by stoppage of the motor, a first memory being non-volatile, and a second memory being volatile and including a cache area, the control method comprising: writing data received from a host device to the one or more magnetic disks via the cache area by using the second power generated by the power supply circuit while the first power is supplied; executing a backup process when supply of the first power is cut off, the backup process being executed by disabling communication with the host device and saving content of the cache area to the first memory by using the third power generated by the power supply circuit; monitoring supply of the first power; keeping connection with the external power supply in a state where the supply of the first power is cut off; and enabling communication with the host device in response to determining, by the monitoring, that the supply of the first power is restored.
9. The control method according to claim 8, wherein the external power supply includes a first external power supply serving to supply fourth power as the first power and includes a second external power supply serving to supply fifth power as the first power, the control method further comprising: monitoring supply of the fourth power and supply of the fifth power; keeping connection with the first external power supply and connection with the second external power supply in a state where either one or both of the supply of the fourth power and the supply of the fifth power are cut off; and enabling communication with the host device in response to determining, by the monitoring, that either the cut off supply of the fourth power or the cut off supply of the fifth power is restored.
10. The control method according to claim 9, wherein the magnetic disk apparatus further including a first electronic fuse and a second electronic fuse, the first electronic fuse being connected between the first external power supply and the power supply circuit, the first electronic fuse serving to enable, by being turned on, supply of the fourth power from the first external power supply to the power supply circuit and serving to cut off the supply of the fourth power by being turned off, the second electronic fuse being connected between the second external power supply and the power supply circuit, the second electronic fuse serving to enable, by being turned on, supply of the fifth power from the second external power supply to the power supply circuit and serving to cut off the supply of the fifth power by being turned off, and the control method further comprising turning on the first electronic fuse and the second electronic fuse in a state where either one or both of the supply of the fourth power and the supply of the fifth power are cut off.

11. The control method according to claim 8, further comprising receiving write data transmitted from the host device and storing the received write data in a first memory when communication with the host device is enabled during execution of the backup process.
  12. The control method according to claim 8, further comprising transmitting information about a state of the magnetic disk apparatus to the host device when communication with the host device is enabled during execution of the backup process.
  13. The control method according to claim 12, wherein the information about the state of the magnetic disk apparatus includes a progress status of the backup process and/or an operation mode of the magnetic disk apparatus.
  14. The control method according to claim 12, further comprising terminating the backup process without waiting for a timeout of the backup process in response to receiving an instruction to interrupt the backup process from the host device after transmitting the information about the state of the magnetic disk apparatus to the host device.
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